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AlphaNET®
Installation Guide

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CHAPTER 1

INSTALLING THE AlphaNET SOFTWARE

Much of the AlphaNET software is the same no matter what physical method of interconnecting your systems you choose to use. This chapter describes how to install that common body of software prior to performing any hardware specific installation which is covered in later sections. In this chapter, we discuss:

- Loading the AlphaNET Software onto Your System
- Creating the SYSNAM.DAT File
- Modifying the System Initialization Command File
- Testing the New System Initialization Command File

1.1 OVERVIEW OF THE SOFTWARE INSTALLATION

Before installing the AlphaNET software, check the following items:

- Make sure your operating system meets the minimum revision level requirements as listed in the software release notes shipped with your AlphaNET Software. Attempts to use AlphaNET with operating system software releases other than those listed in the release notes will yield unpredictable results.
- Make sure consistent versions of the AlphaNET software are installed on all computers on your network. Mixing versions of AlphaNET software will not work.

1.2 LOADING THE ALPHANET SOFTWARE

After logging into DSK0:[1,2], use the following command to copy the software from the tape to your system:

```
VCRRES DSK0:[]=ALL:[] RETURN
```

This will install all necessary software into the correct accounts on your system. Run the VERIFY program to make sure all of the files copied correctly. The file NET.DIR that comes with the release contains a complete list of the software. Enter:

```
VERIFY NET.DIR RETURN
```

1.3 CREATING THE SYSNAM.DAT FILE

Log into DSK0:[1,4] (SYS:), and use AlphaVUE to create a file named SYSNAM.DAT. This file contains only a single line of data—a name of up to thirty characters describing your system. This name is displayed in the status command NTSTAT, and by other commands, and helps other systems on the network know who you are. For example:

```
Accounting System
```

If you do not create a SYSNAM.DAT file, NTSTAT will display a blank for your system description. The release contains an example file named SYSNAM.NEW.

1.4 UPDATING THE SYSTEM INITIALIZATION FILE

Installing the AlphaNET software consists of adding the following items to your system initialization command file:

- An increased number of available jobs to allow for virtual terminal connections and remote procedure calls.
- A background job to run the network service program, NETSER.
- A background job to run the virtual terminal service program, VTSER.
- A MSGINI command (specified **before** the first SYSTEM command) which will allocate memory for AlphaNET's message buffers.
- A SYSTEM command which loads and initializes the remote procedure call handler, RPC.SYS.

AlphaNET version 2.2 users, who are using Ethernet, and operating with AMOS 2.2 (or later), may wish to create a special Ethernet communication "zone." This zone, used in conjunction with the virtual terminal routine CONECT, is allocated via a special SYSTEM statement in the system initialization command file. See the *AlphaNET User's Guide* for information on the purpose and use of the communication zone.

1.4.1 Installing the Product Installation Code

After copying the AlphaNET software onto your system, you must code that software to run on your specific system. To do so, execute the NETINI program from AMOS command level. The first time you do this you will be prompted for a Product Installation Code (PIC).

This PIC is a unique indentifier for your system that must be obtained from your dealer. Enter the PIC, carefully verifying you have entered it correctly and press RETURN.

After a brief pause, you should be returned to AMOS command level and you can proceed with the remainder of the installation. If you see the error message `?Improper SSD,` verify you entered the correct PIC by re-installing the software and trying again. If you still receive the same error, check with your dealer to make sure the correct PIC was supplied for your system.

You must successfully complete this portion of the AlphaNET installation for any other portion to work.

1.4.2 Modifying and Testing the TEST.INI File

Before doing anything else, make a copy of your system initialization command file. We will modify this copy during the installation to make sure any errors or problems that crop up will not affect your ability to boot your system.

To make a copy, log into DSK0:[1,4] and make a copy of your system initialization file. For example:

`COPY TEST.INI=AMOSL.INI` RETURN

or

`COPY TEST.INI=AMOS32.INI` RETURN

After creating TEST.INI, edit it with AlphaVUE and make the changes described in the following sections. Note some of these changes may already have been made, depending on what other software exists on your computer.

1.4.3 Increasing the Number of Available Jobs

AlphaNET makes use of additional jobs during its execution of both virtual terminal connections and remote procedure calls. For this reason, you must increase the maximum number of jobs supported by your system.

Within the TEST.INI file, specify the maximum number of jobs using the JOBS statement located at the beginning of the TEST.INI file. The number following the JOBS command specifies the total number of jobs that can exist on your system at any one

time. You will need to allocate one entry for each of the jobs you are predefining, plus one entry for each simultaneous virtual terminal connection you will have active, plus one entry for each active remote procedure call. Each entry you reserve consumes only 4 bytes of system memory, so it is best to be on the safe side and allocate more JOBS than you think you actually need.

1.4.4 Adding the NETSER Background Job

AlphaNET requires one job be running the NETSER program in the background to act as a server for other systems. You will need to set up such a job, log it into DSK0:[1,4] and have it run NETSER. Because it is a background job, you can use a PSEUDO, NULL terminal to attach to the job.

NETSER needs memory to do its remote procedure calls and name registration functions. The exact memory requirements depends on the operations to be performed in your particular installation. There is no defined minimum or maximum value. We recommend a memory partition of at least 100K.

1.4.5 Adding the VTSER Background Job

A background job is also required to execute VTSER, the virtual terminal server program. If you do not wish to have other systems connect virtual terminals to your system, you do not need to set up this job. Because it is a background job, you can attach a PSEUDO, NULL terminal to it.

VTSER allocates the memory for its virtual terminal connections out of its own partition. For this reason, you must allocate sufficient memory to VTSER for its own needs plus the needs of as many simultaneous virtual terminal connections as you wish to support. A minimum of 128K should be allocated, with 512K or more preferred.

You must make sure NETSER is fully up and running before trying to execute the VTSER program. The best way to do this is to issue a WAIT NETSER command prior to invoking VTSER.

1.4.6 VTSER and Network Security

Users connected to your computer via the virtual terminal server program, VTSER, have essentially the same operational capabilities as direct users. Your system operator should take steps to insure that only authorized persons have access to your computer over the network.

To restrict access to your system by virtual terminals, you can specify a system access password to be requested each time someone tries to connect to your system with a virtual terminal. This password is in addition to the normal login password.

The password is entered immediately following the VTSER command located in the VTSER job initialization file VTSER.JIN. The file will look something like this:

```
:T
LOG DSK0:[1,4]
VTSER MAGICWORD
```

The password, which appears in bold type in the example, can be up to 10 characters long; VTSER does not distinguish between upper and lower case letters. To protect your VTSER password, the file containing the password should be located on a logical disk device which can't be accessed over the network. For example, take a look at the SETJOB command for the VTSER job shown below:

```
SETJOB VTSER,VTSER,512K,VTSER.JIN
```

The system initialization command file with the above VTSER SETJOB statement should also contain the command `SET NOACCESS DSK0:`. If you do not prevent access to DSK0: **anyone** could read your VTSER password over the network and then make a virtual terminal connection to your computer. You may also choose to locate your VTSER.JIN file on some logical disk device other than DSK0:. For example:

```
SETJOB VTSER,VTSER,512K,DSK11:VTSER.JIN[200,1]
```

For best protection from unauthorized virtual terminal connections make sure you SET NOACCESS to the logical device that contains the VTSER.JIN file.

If you are using AMOS 2.2 (or later) software releases with extended disk format, you can create or modify files with user privileges that prevent them from being read over the network. See your AMOS 2.2 documentation for information on creating files with user privileges.

1.4.7 Allocating Message Buffers with MSGINI

AMOS's Intertask Communication System, the foundation for all of AlphaNET, requires buffer space within the system communication area. This space is allocated by the MSGINI command. This command must be performed before the first SYSTEM command, and takes as an argument the amount of memory to allocated. A minimum of 16K should be allocated when using AlphaNET.

1.4.8 Initializing the Remote Procedure Call System

The remote procedure call facility requires you allocate and initialize a support module, RPC.SYS, within system memory. To do so, add the command SYSTEM RPC.SYS/N to the other SYSTEM commands within the system initialization command file.

1.4.9 Defining a Network

For each network your system connects to, you must define the network using the NETINI command. The NETINI command must appear in your system initialization command file after the MSGINI command, but prior to any SYSTEM commands.

The NETINI command takes as an argument the name of a network initialization command file. This network initialization file defines to the system the unique settings and parameters of a network. The exact contents of this file will therefore differ from one type of network to another. However, much of the content is the same regardless of network type and is explained below.

Here's the format of a network initialization command file:

```
DRIVER = xxxxxx
NETWORK = x
GROUP = x
NODE = x
NODECHECK = xxx
NODECOUNT = xx
GROUPCOUNT = xx
NAME = Descriptive Name
```

To create a network initialization file, specify the network driver name for your specific network type in the DRIVER statement. Then specify the network, group, and node numbers identifying your system as a node on a specific network. The network number can be from 1 to 254. Each line of the system initialization command file can be up to 512 characters in length.

The NODECHECK statement tells AMOS whether or not to check for the existence of a node before trying to communicate with it. If NODECHECK is set to ON, you will get an immediate error message (`Node does not exist`) if the node you are trying to communicate with is not found. If NODECHECK is set to OFF, or if the NODECHECK statement is not included in your network initialization file, you will still get the error message, but it will take several seconds longer.

The value defined in the NODECOUNT statement must be equal to, or greater than, the number of nodes on your network. If you do not include a NODECOUNT statement in your network initialization file, the NODECOUNT number defaults to 10. This is not a problem if you have 10 or less nodes defined on your network, but if you have more than 10 nodes defined, you will not be able to communicate with all your nodes and you will assorted network problems.

The value defined in the GROUPCOUNT statement must be equal to, or greater than, the number of groups defined on your network. If you do not include a GROUPCOUNT statement in your network initialization file, the GROUPCOUNT number defaults to 1. This is not a problem if you only have one group defined on your network, but if you have more than one group defined, you will not be able to communicate with all the groups defined on your network.

The NAME statement specifies a symbolic name for each network you define; the NAME statement must not exceed 20 characters.

Using the specific data about your node and network, use AlphaVUE to create a network initialization file.

1.4.10 Defining Ersatz Devices for Your Network

Dealing directly with AlphaNET internet addresses in the form of CPU IDs can be very cumbersome. While these eight to eleven digit numbers serve to uniquely identify each node on all networks, they are certainly not intuitive for the normal user.

Ersatz devices provide a simple way to define a symbolic equivalent to the CPU ID numbers. Anyone wishing to refer to a specific node on the network can then use the symbolic name, rather than the numeric identifier.

Ersatz devices are defined by the ERSATZ command in the system initialization command file. The ERSATZ command reads in a text file defining the ersatz device names and their equivalent CPU IDs and stores them for later use. This text file need only be created once, and then copied to all nodes on the network.

We recommend you create a separate ersatz device definition file for each network on your system.

For example, if the network you are installing is network 1, create the ersatz device definition file VID001.ERZ. The contents of this file, created with AlphaVUE, might look something like:

```
MAIN: = 16842753-
PRINT: = 16842754-
WRHSE: = 16842755-
PAYROL: = 16842757-
ACCT2: = WRHSE:DSK2:[100,4]
```

You can use names up to six characters in length (plus the colon). The ersatz definition can contain other ersatz names, and can specify any part of a complete file specification.

The ERSATZ command may be used multiple times within the system initialization command file to process multiple ersatz device specification files. All of the ERSATZ commands, however, must be grouped together, with no intervening commands.

1.4.11 A Sample Initialization File

This sample system initialization command file is intended to give you a better understanding of how to set up each of the previously discussed commands. Of course, the details, such as number and type of devices, non-AlphaNET jobs, etc., will differ with your own system; this example simply serves to illustrate the AlphaNET portions of the system initialization command file. The portions shown in bold are the AlphaNET related sections.

```
:T
JOBS 50
JOBALC JOB1,NETSER,VTSER
TRMDEF TERM1,AM1200=0:9600,ALPHA,100,100,100
TRMDEF NETSER,PSEUDO,NULL,100,100,100
TRMDEF VTSER,PSEUDO,NULL,100,100,100
DEVTBL DSK1
DEVTBL TRM,RES,MEM,/VCR
BITMAP DSK,608,0,1
MSGINI 16K                  ; Allocate memory to ITC
NETINI VID001.INI           ; Initialize a network
ERSATZ ERSATZ.INI           ; Set up ersatz devices
ERSATZ VID001.ERZ           ; Set up net ersatz devices
SYSTEM SYSMSG.USA
SYSTEM RPC.SYS/N            ; Initialize RPC
SYSTEM                      ; Ends module allocation
SETJOB NETSER,NETSER,100K,NETSER.JIN
WAIT NETSER
SETJOB VTSER,VTSER,512K,VTSER.JIN
MEMORY 0
```

The job initialization file NETSER.JIN should contain the following:

```
:T
LOG DSK0:[1,4]
NETSER
```

The job initialization file VTSER.JIN should contain the following:

```
:T
LOG DSK0:[1,4]
VTSER {password}
```

If your system uses user names, you will need to add the appropriate user names and passwords, if needed, to the job initialization files.

1.5 CONECT SUPPORT (ETHERNET ONLY / AMOS 2.2 ONLY / ALPHANET 2.2 ONLY)

If you are running AlphaNET 2.2, using an AMOS 2.2 (or later) operating system, and using Ethernet as your network data transport device, there are some additional features built into the terminal connection program CONECT. These features include an improvement in the virtual terminal performance, 8-bit character support, and the availability of the command line editor.

To use these new features under CONECT, you will need to add a special TRMDEF statement. You must also create a special communication "zone," which requires one additional statement in your system initialization command file.

If you are using a serial or video network, or if you not using AlphaNET 2.2 and AMOS 2.2, the following two sections are not applicable to your configuration.

1.5.1 TRMDEF Statement (CONECT Support Only)

The purpose of this special TRMDEF statement is to allocate a pool of Terminal Control Blocks (TCBs) to be used by jobs "spawned" by VTSER in response to a remote connection request issued by CONECT. Note that this is applicable only when using Ethernet as a network transport device.

The TRMDEF command line syntax for allocating TCBs is:

TRMDEF #count In-width, In-buffer, Out-buffer, Editor=Number of recall buffers

Example:

```
TRMDEF #10 100,100,100,EDITOR=10
```

As I'm sure you noticed, this is similiar to the traditional TRMDEF statement. The "#10" entry causes 10 TCBs with associated terminal buffers to be made available for use by remote job connections.

For best results, the special TRMDEF statement described above should be the last TRMDEF statement in your system initialization command file.

1.5.2 Creating Communication Zone (CONECT Support Only)

In order to support remote connections requested by CONECT, the host computer must have a special communication "zone" defined in the initialization command file. This zone is part of the interface between the Ethernet network driver and the spawned jobs created as the result of a CONECT connection request. To set up this area, an ETHZON statement must be added to the system initialization command file.

The syntax is simply:

ETHZON *maxjobs*

Where *maxjobs* is the maximum number of CONECT connections to be supported at any given time. Note, the "maxjobs" number should match the number listed in the special TRMDEF statement described above.

Example:

```
ETHZON 10
```

This statement, when added to your system initialization command file, would allocate an Ethernet communication zone large enough to support a maximum of 10 virtual terminal connections created by a CONECT request. Make sure you insert the ETHZON statement after the TRMDEF statements and before the the first NETINI command.

There are some restrictions to keep in mind when defining your enhanced virtual terminal connections. See the Section on enhanced virtual terminal restrictions.

1.5.3 Enhanced Virtual Terminal Restrictions

The most important point to keep in mind when defining the enhanced virtual terminal connections is that they are "real jobs" that count against the licensed number of jobs allocated in your AMOS operating system. Every TRMDEF statement in your system initialization command file that defines the use of a physical port represents a "real job." Make sure the number of enhanced virtual terminal connections you define in the special TRMDEF and ETHZON statements (when added to your existing real jobs) does not cause you to exceed the number of licensed jobs in your AMOS operating system.

If there are more enhanced virtual terminal connections defined than your AMOS port license supports, **no** TCBs will be allocated for enhanced virtual terminal connections.

The number of jobs defined in the JOBS statement at the top of your system initialization command file must be equal to, or greater than, the total number of jobs you wish to support at any given time. This includes all real jobs, which includes enhanced virtual terminal connections, plus all PSEUDO jobs.

If you define five as the number of enhanced virtual terminal connections for your computer, any virtual terminal connections beyond five will use the standard non-enhanced version of CONECT.

1.5.4 TDVDEF (AMOS 2.2 only)

The TDVDEF statement (valid only with AMOS 2.2 or later releases) must be inserted into your system initialization command file after the last TRMDEF statement and before the first NETINI command. Using the TDVDEF statement, you can load any or all of your terminal driver files into memory when your computer boots. You can follow the TDVDEF statement with the wild card symbol (*) and all the TDV files in DSK0:[1,6] will be loaded into memory, or you can follow the TDVDEF statement with specific TDV names:

```
TDVDEF *
or
TDVDEF AM65.TDV, AM72.TDV, AM62A.TDV
```

1.5.5 Testing the New Initialization Command File

After making the changes described above, log into DSK0:[1,2] and test your new initialization command file using the MONTST command. For AMOS/L systems, enter:

MONTST AMOSL.MON,TEST.INI RETURN

For AMOS/32 systems, enter:

MONTST AMOS32.MON,TEST.INI RETURN

If the system doesn't reboot properly, press your computer's RESET button to reboot under your original initialization file. Re-check TEST.INI for mistakes and try again.

Once the system has booted correctly, use SYSTAT to make sure the job running NETSER is in an MS state, and the job running VTSER is in an SL state. If not, recheck your TEST.INI file to make sure these jobs were properly initialized. If you are unsuccessful after several tries, you may wish to attach either one of these background jobs to a real terminal during the boot process so you may see any error messages they may be reporting.

When you have successfully completed and MONTSTed the TEST.INI file, rename it to replace your previous initialization command file. On an AMOS/L system, enter:

COPY AMOSL.OLD=AMOSL.INI RETURN
RENAME/D AMOSL.INI=TEST.INI RETURN

On an AMOS/32 system, enter:

COPY AMOS32.OLD=AMOS32.INI RETURN
RENAME/D AMOS32.INI=TEST.INI RETURN

You can now install the physical network connection and its related software. This installation is described in the following sections.

1.6 DETERMINING YOUR CPU ID NUMBER

If you wish to find out what your cpuID number will be (which will save you time if you want to define ersatz names), you can use the AlphaBASIC program listed on the next page. It asks you for information from your network initialization file, and returns the cpuID AlphaNET will give you.

```
MAP1  CPU'SOURCE,B,4
MAP1  CPU'NUM,@CPU'SOURCE
      MAP2  NET'NODE,B,2
      MAP2  NET'GROUP,B,1
      MAP2  NET'NUM,B,1
MAP1  ANSWER,S,1

START:
  INPUT LINE "Enter your Network number: ";NET'NUM : PRINT
  INPUT LINE "Enter your Group number: ";NET'GROUP : PRINT
  INPUT LINE "Enter your Node number: ";NET'NODE : PRINT
  PRINT "Your cpuID number is: ";
  PRINT CPU'SOURCE USING "##########";"-" : PRINT
  INPUT "Would you like to find another cpuID (Y/N)? ";ANSWER
  IF UCS(ANSWER) = "Y" THEN GOTO START
END
```

CHAPTER 2

INSTALLING NETWORK GATEWAYS

Gateways allow you to gain access to networks your system is not physically connected to. This is done by routing all traffic for that network to a network node which does have a physical connection. This extra routing is transparent to both you and the software in use. You must, however, pre-define all networks to which you wish to connect, and establish routing paths for each.

You define a gateway much as you would any other network, using the NETINI program, except instead of defining a physical connection by specifying a network driver, you specify a logical connection by specifying the network node to forward messages to.

The elements required to connect different networks through a common gateway are illustrated on the next two pages. In Figure 2-1 four networks are shown. Networks #1, #2, and #3 are serial networks. A serial network is a point-to-point type network. This means each point to point connection makes up a unique network. In Figure 2-1 you can see Network #1 connects Computer A to Computer E; Network #2 connects Computer B to Computer E; Network #3 connects Computer C to Computer E.

Network# 4 is a bus topology type network that allows a large number of systems to share a common network. Computer D, Computer E, and Computer F are able to communicate with each other, because they are all part of a common network (Network #4).

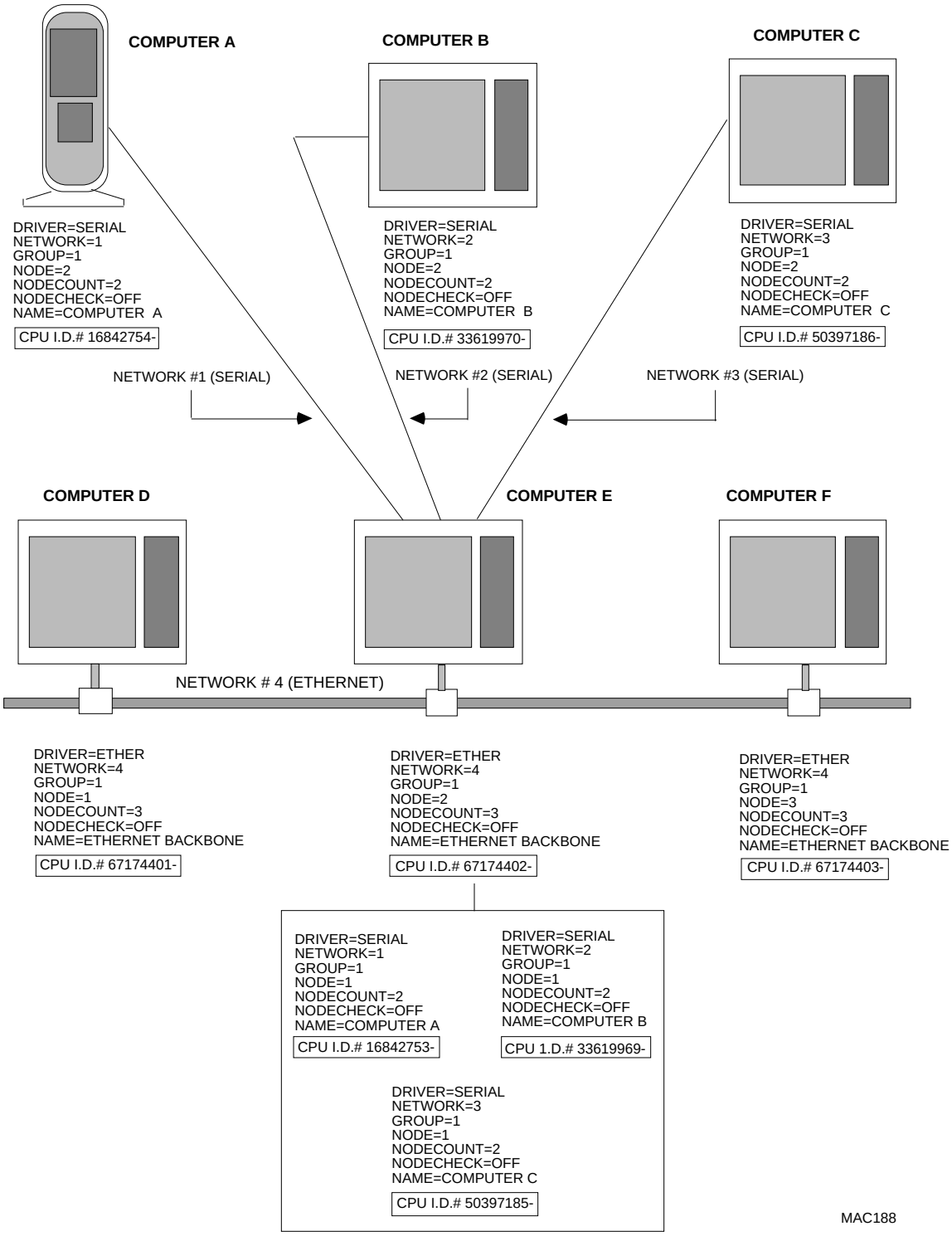

COMPUTER A
COMPUTER B
COMPUTER C
DRIVER=SERIAL
NETWORK=1
GROUP=1
NODE=2
NODECOUNT=2
NODECHECK=OFF
NAME=COMPUTER A
CPU I.D.# 16842754-
DRIVER=SERIAL
NETWORK=2
GROUP=1
NODE=2
NODECOUNT=2
NODECHECK=OFF
NAME=COMPUTER B
CPU I.D.# 33619970-
DRIVER=SERIAL
NETWORK=3
GROUP=1
NODE=2
NODECOUNT=2
NODECHECK=OFF
NAME=COMPUTER C
CPU I.D.# 50397186-
NETWORK #1 (SERIAL)
NETWORK #2 (SERIAL)
NETWORK #3 (SERIAL)
COMPUTER D
COMPUTER E
COMPUTER F
NETWORK # 4 (ETHERNET)
DRIVER=ETHER
NETWORK=4
GROUP=1
NODE=1
NODECOUNT=3
NODECHECK=OFF
NAME=ETHERNET BACKBONE
CPU I.D.# 67174401-
DRIVER=ETHER
NETWORK=4
GROUP=1
NODE=2
NODECOUNT=3
NODECHECK=OFF
NAME=ETHERNET BACKBONE
CPU I.D.# 67174402-
DRIVER=ETHER
NETWORK=4
GROUP=1
NODE=3
NODECOUNT=3
NODECHECK=OFF
NAME=ETHERNET BACKBONE
CPU I.D.# 67174403-
DRIVER=SERIAL
NETWORK=1
GROUP=1
NODE=1
NODECOUNT=2
NODECHECK=OFF
NAME=COMPUTER A
CPU I.D.# 16842753-
DRIVER=SERIAL
NETWORK=2
GROUP=1
NODE=1
NODECOUNT=2
NODECHECK=OFF
NAME=COMPUTER B
CPU 1.D.# 33619969-
DRIVER=SERIAL
NETWORK=3
GROUP=1
NODE=1
NODECOUNT=2
NODECHECK=OFF
NAME=COMPUTER C
CPU I.D.# 50397185-
MAC188

Figure 2-1

Look at the network example in Figure 2-1; note there are network initialization files defining each system (node) on all four networks. However, the elements necessary for creating a gateway between the networks are missing. Without the gateway defined, the communication between systems in our network example would be very limited. Each computer would only be able to communicate with other computers within its local network:

- Computer A would not be able to communicate with Computers B, C, D, and F.
- Computer B would not be able to communicate with Computers A, C, D, and F.
- Computer C would not be able to communicate with Computers A, B, D, and F.
- Computer D would not be able to communicate with Computers A, B, and C.
- Computer E is directly attached to all the computers, therefore it is able to communicate with each system on all four networks.
- Computer F would not be able to communicate with Computers A, B, and C.

By defining gateways, we can fully connect this network. To define a gateway, you create an initialization file (filename.GIN) for NETINI containing a NETWORK= statement defining the network you wish to make available, a NAME= statement to assign a symbolic name, and a FORWARDTO= statement to define the network address of the node to which to forward all traffic for the network being defined.

To fully interconnect all six systems shown in Figure 2-1, each of the systems would have to define gateways as shown in Table 2-1 (in addition to defining the networks to which they are physically connected):

The network initialization files (*.NIN) must be defined in your system initialization command file before your gateway initialization files (*.GIN).

NETWORK GATEWAYS

COMPUTER A

```
DRIVER=SERIAL
NETWORK=1
GROUP=1
NODE=2
NODECOUNT=2
NODECHECK=OFF
NAME=COMPUTER  A
```

```
NETWORK=4
NODECHECK=OFF
NAME=ETHERNET BACKBONE
FORWARDTO=16842753-
```

```
NETWORK=2
NODECHECK=OFF
NAME=COMPUTER  B
FORWARDTO=16842753-
```

```
NETWORK=3
NODECHECK=OFF
NAME=COMPUTER C
FORWARDTO=16842753-
```

COMPUTER B

```
DRIVER=SERIAL
NETWORK=2
GROUP=1
NODE=2
NODECOUNT=2
NODECHECK=OFF
NAME=COMPUTER  B
```

```
NETWORK=4
NODECHECK=OFF
NAME=ETHERNET BACKBONE
FORWARDTO=33619969-
```

```
NETWORK=1
NODECHECK=OFF
NAME=COMPUTER  A
FORWARDTO=33619969-
```

```
NETWORK=3
NODECHECK=OFF
NAME=COMPUTER C
FORWARDTO=33619969-
```

COMPUTER C

```
DRIVER=SERIAL
NETWORK=3
GROUP=1
NODE=2
NODECOUNT=2
NODECHECK=OFF
NAME=COMPUTER  C
```

```
NETWORK=4
NODECHECK=OFF
NAME=ETHERNET BACKBONE
FORWARDTO=50397185-
```

```
NETWORK=1
NODECHECK=OFF
NAME=COMPUTER  A
FORWARDTO=50397185-
```

```
NETWORK=2
NODECHECK=OFF
NAME=COMPUTER B
FORWARDTO=50397185-
```

COMPUTER D

```
DRIVER=ETHER
NETWORK=4
GROUP=1
NODE=1
NODECOUNT=3
NODECHECK=OFF
NAME=ETHERNET BACKBONE
```

```
NETWORK=1
NODECHECK=OFF
NAME=COMPUTER A
FORWARDTO=67174402-
```

```
NETWORK=2
NODECHECK=OFF
NAME=COMPUTER B
FORWARDTO=67174402-
```

```
NETWORK=3
NODECHECK=OFF
NAME=COMPUTER C
FORWARDTO=67174402-
```

COMPUTER E

```
DRIVER=ETHER
NETWORK=4
GROUP=1
NODE=2
NODECOUNT=3
NODECHECK=OFF
NAME=ETHERNET BACKBONE
```

```
DRIVER=SERIAL
NETWORK=1
GROUP=1
NODE=1
NODECOUNT=2
NODECHECK=OFF
NAME=COMPUTER A
```

```
DRIVER=SERIAL
NETWORK=2
GROUP=1
NODE=1
NODECOUNT=2
NODECHECK=OFF
NAME=COMPUTER B
```

```
DRIVER=SERIAL
NETWORK=3
GROUP=1
NODE=1
NODECOUNT=2
NODECHECK=OFF
NAME=COMPUTER C
```

COMPUTER F

```
DRIVER=ETHER
NETWORK=4
GROUP=1
NODE=3
NODECOUNT=3
NODECHECK=OFF
NAME=ETHERNET BACKBONE
```

```
NETWORK=1
NODECHECK=OFF
NAME=COMPUTER A
FORWARDTO=67174402-
```

```
NETWORK=2
NODECHECK=OFF
NAME=COMPUTER B
FORWARDTO=67174402-
```

```
NETWORK=3
NODECHECK=OFF
NAME=COMPUTER C
FORWARDTO=67174402-
```

MAC189

Table 2-1

When actually defining a FORWARDTO address, you may use an ERSATZ name in place of the CPU I.D. number (that's the big decimal number). See Chapter 1 of this document for more information on the use of ERSATZ names.

CHAPTER 3

INSTALLING VideoNET

This chapter gives you an overview of the VideoNET hardware, and an explanation of how VideoNET works. VideoNET is one of several data transport mechanisms that works in conjunction with the AlphaNET software package. VideoNET uses the existing video backup hardware on your computer system.

AM-1000 and S-100 based systems may require a hardware upgrade before being able to use VideoNET. For instructions on how to verify your hardware is up-to-date, and for instructions on how to update the hardware for VideoNET, see *System Upgrade Instructions: VideoNET*, PDI-00615-02.

If you have an AM-1000 or an S-100 based system, check the System History Label (found on a side panel of an S-100 system or the bottom panel of an AM-1000 system)—if your system is not marked "VideoNET," but contains the video backup hardware, you will need one of the VideoNET upgrade kits listed below.

Systems such as the AM-1200, AM-1400, AM-1600, AM-2000M, and VME-bus based systems do not require VideoNET-specific upgrades, but you should nonetheless make sure your hardware is up to the latest recommended revision level before proceeding with the VideoNET installation.

Only systems with video backup capability may be upgraded to use VideoNET. It is not possible to upgrade AM-610 boards earlier than revision level D00.

For each S-100 Bus system not previously upgraded to AlphaNET compatibility, you will need one upgrade kit (PDB-00615-03) for each AM-610 VCR Controller board (Rev. D00-D06). Revisions D07 and later already have the AlphaNET upgrade installed.

For each AM-1000 system not previously upgraded to AlphaNET compatibility, you will need one upgrade kit (PDB-00615-02) for each AM-167 board (Rev. A00 to D18). Revisions D19 and later already have the AlphaNET upgrade installed.

VideoNET physically consists of the VCR interface and appropriate internal cabling to install a connection on the rear panel of the computer system. BNC coaxial connectors are provided at system rear panels. "T" connectors physically tap the node into the network.

The person who installs VideoNET must provide all external cables and "T" connectors that plug into the "LINK I/O" connector on the rear panel of each system.

Note VideoNET is intended for use within the same building structure. Applications requiring interplant connections are outside the scope of this product.

3.1 PRODUCT DESCRIPTION

The basic features and capabilities of VideoNET are:

- Full AlphaNET support using a low-cost, medium-performance link using existing video interfaces.
- Support for interconnection of up to 64 system nodes.

3.2 VIDEONET SPECIFICATIONS

Network type:	Token Bus
Topology:	Bus - Local Area Network (LAN) or a segment, Tree - Network System.
Maximum Segment Length:	1200 ft. - Twinaxial cable (Trunk Cable)
Max. Nodes per LAN:	64
Medium:	Twinaxial cable recommended
Bit Rate:	1 million BPS
Termination:	75 ohm passive terminators at each end of the trunk cable.
Packet Size:	Fixed 512 data bytes within a fixed 563 byte packet.
AC Ground Noise Immunity:	

3.3 UPGRADE KITS

The package you have purchased is determined by the type of Alpha Micro systems you intend to link, and existing system hardware. The Upgrade Kits are required only for systems which are not labeled "VideoNET" on their system history labels.

The "System Upgrade Instructions" document for VideoNET (PDI-00615-02) is included with each kit. There are two upgrade kits for VideoNET. Each kit is for a particular type of system and contains the following pieces:

PDB-00615-02 AM-1000 Systems with VCR backup capability:

The upgrade must be performed on all Rev. A and B configurations, and up to (& including) Rev. D18, of the AM-167 CPU board. Rev. D19, and later, of this board are upgraded at the factory. Physical interface (LINK I/O) connectors already exists on systems with video backup (i.e., AM-1000(E,X) VW, and VWF).

a. Appropriate VideoNET microcode PROM (Part #DWP-00169-05 Rev. B00 or later)

b. An assortment of 8 resistors, a capacitor, and a diode.

PDB-00615-03 AM-100/L Systems:

The upgrade to be performed on Rev. D00-D06 ONLY of the AM-610 VCR Interface board. All later revision levels of this board are upgraded at the factory.

a. Appropriate VideoNET microcode PROM (Part #DWP-00169-05 Rev. B00 or later)

b. Internal network drop cable (Part # DWB-10135-00) with rear panel mounting brackets to adapt cable to older RS-232 cutouts (DWF-20332-01) or newer rear panel slots (DWF-20332-00).

c. An assortment of 8 resistors, a capacitor, and a diode.

3.4 ENVIRONMENTAL SPECIFICATIONS

The Operating Temperature and Humidity Specifications are the same for VideoNET as for the systems VideoNET is used with. Avoid noisy EMI/RFI environments, such as close proximity to power cables, welding machines, and transmitter cables.

Cable shielding should not contact any objects such as structural metal, ducting, plumbing fixtures or other unintentional conductors.

Be aware of any local building codes that will affect your installation.

3.5 VIDEONET HARDWARE INSTALLATION

In this section we discuss the following:

- Finding and setting network terminators.
- Hooking up VideoNET cables.

3.5.1 Network Terminators

All computers are shipped by Alpha Micro have their video network terminator jumpers installed on their respective circuit boards (i.e., S-100/AM-610, AM-1000/AM-167, VME/AM-630, AM-1200/AM-177, AM-1400/AM-130 and AM-134, AM-1600/AM-135, and AM-2000M/AM-145). One of the first steps in the installation process is to remove these network terminator jumpers on all computers which will be connected to your network.

The video network must be terminated at each end of the network, but this termination is done external to the computers.

All computers must be opened to remove the network terminator jumper.

The location of the network terminator jumpers is as follows:

1. S-100 Computers (AM-610 Board) — The terminator jumper (TERM) is located at the top left of the AM-610 board as you face the component side (between U2 and U3, and between the J2 "LINK I/O" and J3 "VIDEO OUT" connectors). Remove the jumper.

2. AM-1000 Computers (AM-167 Board) — The "TERM" jumper is located near the rear of the AM-167 board (revision levels B and D), between the J9 "LINK I/O" and J10 "VIDEO IN" connectors. There are two sets of jumper pins; the TERM set is just forward of the TST set. Remove the jumper.

   The AM-167 Revision A00 board does not have a "TERM" designation for the jumper. The jumper, however, is in approximately the same location as for the other revision levels (between J9 "LINK I/O" and J10 "VIDEO IN"). It is also between the integrated circuit designated "U6" and resistor "R72." Remove the jumper.

3. AM-1200 Computers (AM-177 Board) — The "TERM" jumper on the AM-177 is located near the rear of the board directly behind the "LINK" BNC connector. Remove the jumper.

4. AM-1400/1600 Computers (AM-130/AM-134/AM-135 Boards) — The network terminator jumper for these boards is located near the "LINK" BNC connector. The jumper is labeled as either "TERM" or W3. Remove the jumper.

5. AM-2000M Computers (AM-145 Board) — The terminator jumper (labeled LINK TERM) is located on the right rear of the CPU board (as seen from the front of the computer), near the VideoNET connector. Remove the jumper.

6. All VME Computers (AM-630 Board) — The terminator jumper (labeled W 35) is located at the top of the AM-630 board. Remove the jumper.

If for some reason you are not sure if your network terminator jumper has been removed, there is an easy way to check without having to open the computer enclosure. This procedure is described in the following section.

3.5.2 Network Terminator Test

The following procedure will help you detect computers with installed terminators without having to remove the computer top covers. This procedure requires the use of either a digital or analog ohmmeter; their probes will be inserted into the rear panel "VIDEO LINK" connector.

With a digital ohmmeter:

1. Place the resistance scale of the digital ohmmeter at 200 ohms. Short the two probes together and check the display for zero.

2. Gently place one probe into the center conductor of the BNC connector located on the computer's rear panel. Put the other probe on the outer shell of the BNC connector.

3. If the measured resistance is 75 ohms or less, then the terminator is connected inside the computer.

4. If the resistance is much greater than 75 ohms, switch the scale of the meter to 2K ohm and re-zero the scale. Re-measure. If the meter reads more than 200 ohms on this scale, then the terminator is not connected.

With an analog ohmmeter:

1. Place the meter at the most sensitive range for this measurement. Depending upon the meter, the maximum range would be one just greater than 75 ohms (i.e., Rx1 or Rx10).

2. Gently place one probe into the center conductor of the BNC connector. Put the other probe on the outer shell of the BNC connector. If the result measures 75 ohms or less, the terminator is installed; from 3 times larger than 75 ohms means the terminator is not installed.

3.6 HOOKING UP THE CABLES FOR VIDEONET

The external cabling required to construct a video network is quite simple and easy to install. Only three components are required:

1. BNC "T" Type Connectors (One for each computer attached to the network.)

2. 75 ohm coaxial cable with a BNC connector at each end (Used to link BNC "T" connectors together. The minimum cable lenght for a VIDEOnet installation is 12 feet.

3. Two 75 ohm BNC type terminators (One for each end of the video network.)

Use the following instructions for cabling together your video network:

1. Each computer will have a BNC type connector extending from its rear panel.

2. Attach one BNC "T" type connector to the BNC connector on the rear panel of each computer. The BNC "T" type connector is shown in Figure 3-1.

3. The computers are linked together by connecting the 75 ohm coaxial cables between the BNC "T" type connectors. The end result will be an unbroken chain of computers linked together by BNC "T" type connectors and 75 ohm coaxial cable.

4. The last step is to install the BNC 75 ohm terminators at each end of the network. The 75 ohm BNC terminator is shown in Figure 3-1.

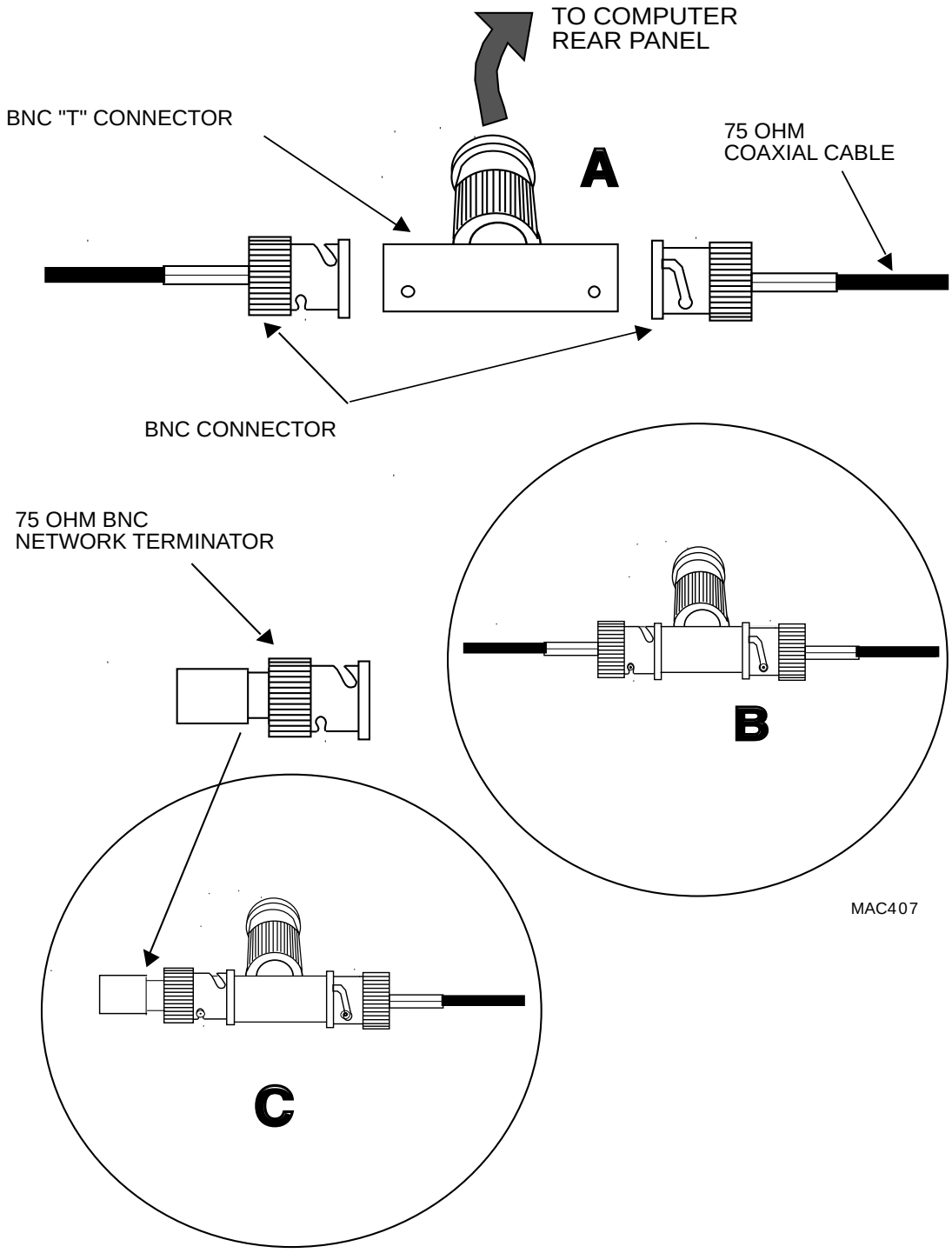


Figure 3-1 -- VideoNET Cabling Components

3.7 VIDEONET SOFTWARE CONFIGURATION

In this section, we discuss:

- Creating a VIDnnn.NIN File. Where "nnn" is the network number.
- Testing the New System Initialization Command File

3.7.1 Overview of the Software Installation

To install the VideoNET software:

- Make sure you have successfully installed the AlphaNET software, as described in Chapter 1 of this book.
- Make certain the VideoNET microcode file 610NET.DVR has been installed into DSK0:[1,6].
- After copying the software onto the system, run the NETINI program and enter the Product Installation Code (PIC).
- Create a network initialization file, VIDnnn.NIN, to install your system onto the network with a unique group and node designation. The network initialization file is also described in Part I.
- Modify your System Initialization Command file to execute the necessary programs to bring up the VideoNET system.

3.7.2 Creating a Network Initialization File for VideoNET

The following is an example of the file (VIDnnn.NIN) that will install your system on the network. It is set up as described in Part I of this book, with certain options set to accommodate your system with unique group and node numbers as well as the correct driver for your VideoNET hardware configuration.

The sample below shows a typical network initialization file for VideoNET. You must specify the correct VideoNET driver in the DRIVER statement, the correct network, group, and node numbers, and must turn NODECHECK on. If you have an S-100 based computer, use the VID100.DVR driver. For any other type of system, use the VIDNDV.DVR driver. Note the VIDVME.DVR driver once used by AlphaNET is no longer functional—if you already have a network initialization file that uses the VIDVME driver, change it to VIDNDV.

```
DRIVER = VIDNDV
NETWORK = 1
GROUP = 1
NODE = 3                      ; Must be unique within group
NODECOUNT = 35                ; Set > number of nodes for
                              ;   future expansion
GROUPCOUNT = 5                ; = > than the number of groups defined
NODECHECK = ON                ; Checks node # before request
NAME = Video Network          ; Descriptive name
```

The NETWORK, GROUP, and NODE numbers define your computer's place in the network. Each type of network has its own unique network number (for example, if you have a video network and a serial network, the video network might be network 1, and the serial network number 2). On each network, the various computers connected to it are identified by the group and node numbers. Your computer might be node 3 in group 1 (as in the example above).

Using the specific data about your place in the network, use AlphaVUE to create a VIDnnn.NIN file. Finish out of the file, and your VIDnnn.NIN file will be ready.

3.7.3 Setting Up Ersatz Names

You will probably want to set up ersatz device names for the nodes on your VideoNET, as described in Chapter 1. By using ersatz device names, you can avoid the use of lengthy and cumbersome network address numbers.

3.7.4 Checking Out the Installation

Once you have made all of the changes to your system initialization command file and the network initialization command file, you are ready to start testing. As a final checklist, make sure you have done the following:

1. Installed the correct version of the microcode (610NET.DVR[1,6]).
2. Installed the AlphaNET software (including installing the PIC) and modified your system initialization command file to include MSGINI, NETSER, VTSER, and a NETINI statement.
3. Created a VIDEO.INI file to define your system as a VideoNET node.

To test:

1. Boot the system by MONTST'ing the new TEST.INI file.
2. If the boot up was successful, rename TEST.INI to AMOSL.INI or AMOS32.INI.

3. Physically attach the system to the network (only after successfully booting the system).

4. Boot up the system using the new system initialization command file.

5. Use the SET LINK command to connect your system to the network. Enter your cpuID number after SET LINK. For example:

```
SET LINK 21569483- RETURN
```

6. Use NTSTAT to see a list of all other nodes on the network.

7. If the network does not come up, recheck all items in this list. It is also possible for the cable or a connector to be bad.

8. If, after the network is up, you wish to do a VCR backup or certify a video tape, you must first disconnect your system from the network. Enter SET NOLINK and your cpuID. You may then use the VCR backup software. To connect to the network again, use SET LINK as shown in step five.

When you have successfully completed TEST.INI, and MONTSTed it, you may rename the TEST.INI to your regular system initialization command file name. For AMOS/L systems, enter:

```
COPY AMOSL.OLD=AMOSL.INI RETURN
RENAME AMOSL.INI=TEST.INI/D RETURN
```

For AMOS/32 systems, enter:

```
COPY AMOS32.OLD=AMOS32.INI RETURN
RENAME AMOS32.INI=TEST.INI/D RETURN
```

3.8 TESTING YOUR VIDEONET INSTALLATION

This section describes how to verify that your VideoNET hardware is operational. The following test should not be run until you have completed all the hardware and software installation instructions described earlier in this chapter.

3.8.1 Using the VNETCK Command

The VNETCK command certifies the video hardware being used by your network is working properly. It is much like the CRT610 command used with the VCR backup software. In order to use VNETCK, your system must be OFF the network. For example:

```
SET NOLINK 1684527- RETURN
VNETCK RETURN
```

and you will see something like:

```
AlphaNET Certify X.X(xxx)
CHECK
Reading Network
Total Blocks Read - 0
Total Copies Read - 1
Total CRC Errors  - 0
Total Hard Errors - 0
Reliability Ratio - 0:1

Certification is complete
```

See the CRT610 reference sheet in your *System Commands Reference Manual* for information on the VNETCK display.

There is one option, /V, that can be used with VNETCK in special circumstances to get a continuous verification of the network. The following steps MUST be followed in order to use /V:

1. All of the nodes MUST be OFF the network.

2. ONE node (the node you suspect might have a problem) enters:

   VNETCK/V RETURN

3. Next the program prompts you to enter the number of hours, enter 1 or more and press RETURN.

4. The program will ask you to enter the number of copies you want, enter 2 or more and press RETURN.

5. The program will begin to run.

Do not bring any system up on the network while VNETCK/V is running.

6. Any of the nodes that wish to see the display can run VNETCK (without /V). Those nodes will then see a continuous display of information about the node that is using /V.

CHAPTER 4

Installing SerialNET

This chapter gives you an overview of the hardware used by SerialNET. SerialNET uses the existing serial terminal interface hardware on your computer system.

Due to the availability of serial interfaces on every Alpha Micro computer, SerialNET makes an ideal interconnection for setting up gateways between networks.

Because Alpha Micro has used a wide variety of serial interfaces over the years, there are some differences in capabilities and performance between these different interfaces when used with SerialNET. In particular, not all of the interfaces support hardware handshaking. Those interfaces which fully support SerialNET with hardware handshaking include:

AM-1003	AM-1000 Expansion Ports (Non-Z80 controlled)
AM-1006	AM-1000 Expansion Ports
AM-1203	AM-1200 Expansion Ports
AM-355	AM-1500/AM-2000 Expansion Ports
AM-130/134	AM-1400 Expansion Ports
AM-135	AM-1600 Expansion Ports
AM-145	AM-2000M Expansion Ports
AM-185	AM-3000 Expansion Ports
AM-140	AM-3000M Serial Ports
AM-355B	AM-3000M Expansion Ports

By using hardware handshaking, speeds up to 57.6 Kbaud is supported. Other interfaces without hardware handshaking are not recommended for use with SerialNET.

The AM-310, AM-1013, and AM-1213 boards are not recommended for use with SerialNET. While the AM-350 board is very useful for speeding up system throughput, it is not the place to connect the SerialNET to—use an AM-355 port off the CPU board.

4.1 PRODUCT DESCRIPTION

The basic features and capabilities of SerialNET are:

- Full AlphaNET support using a low-cost, point-to-point link.
- Support for communication speeds up to 57.6 KBaud.
- Multiple SerialNET connections can be made to a single system.

4.1.1 SerialNET Specifications

Network type:	Point-to-point asynchronous connection
Topology:	Point-to-point
Maximum Segment Length:	50 ft. (RS-232), 1000 ft. (RS-422)
Max. Nodes/Net:	2
Medium:	Shielded cable recommended
Bit Rate:	Up to 57,600 BPS
Termination:	Not needed
Packet Size:	12 to 4096 bytes per packet

4.1.2 Building Cables for SerialNET

The cables used by SerialNET are constructed in the same manner as normal terminal cables, including shielding and length considerations. The connections that must be made are:

RS-232 connection (DB-25 or DB-9):

```
2 <------ 3
7 <-----> 7
3 <-----> 2
4 <------ 5
5 ------> 4
```

RS-422 connection (DB-9):

```
2 <------ 3
8 <------ 6
3 ------> 2
6 ------> 8
4 <------ 5
5 ------> 4
7 <-----> 7
```

4.2 INSTALLING THE SERIALNET HARDWARE

In this chapter, we discuss:

- Creating a SerialNET cable
- Creating a Network Initialization file
- Modifying the System Initialization Command File
- Testing the new System Initialization Command File

4.2.1 Overview of the Software Installation

You must do the following to install the SerialNET software:

- Make sure you have successfully installed the AlphaNET software, as described in Part I of this book.
- After copying the software onto the system, run the NETINI program and enter the Product Installation Code (PIC).
- Create a unique network initialization file, SERnnn.NIN, to install your system onto the network with a unique group and node designation. The network initialization file is also described in Part I.
- Modify your System Initialization Command file to define the serial interface connection you wish to use, and to execute the necessary programs to bring up the SerialNET system.

DO NOT modify your system initialization command file directly, but indirectly by creating a test copy, TEST.INI, and modifying that. Once this copy is executing properly, you may replace your original system initialization command file with the test copy.

4.2.2 Creating a Network Initialization File for SerialNET

The following is an example of the file (SERnnn.NIN) that will install your system on the network. Since the network number is defined as 3, well will create a file called SER003.NIN. It is set up as described in Chapter 1 of this book, with certain options (see the example below) set to accommodate your system. This includes a network, group and node number as well as the SerialNET driver program. The network must be the unique number of the SerialNET, and the node number must be unique within the group your computer is in.

The sample below shows a typical network initialization file for SerialNET. Note you must specify the correct network, group, and node numbers, and must turn NODECHECK off.

```
DRIVER = SERIAL          ; Network driver
NETWORK = 3              ; Network number
GROUP = 1                ; Group number
NODE = 1                 ; Node number (unique
                         ;   within a group and network)
NODECHECK = OFF          ; Checks node # before request
NAME = Serial Network    ; Descriptive name
```

The NETWORK, GROUP, and NODE numbers define your computer's place in the network. Each type of network has its own unique network number (for example, if you have a video network and a serial network, the video network might be network 1, and the serial network number 2). On each network, the various computers connected to it are identified by the group and node numbers. Your computer might be node 3 in group 1 (as in the example above).

Using the specific data about your place in the network, use AlphaVUE to create a SER003.NIN file. Finish out of the file, and your SER003.NIN file will be ready.

4.2.3 Defining the SerialNET Terminal Connection

SerialNET uses a standard terminal port as its physical connection. This terminal port must be defined to AMOS through the use of the TRMDEF statement in the system initialization command file, the same as for any other terminal.

The only unique thing about defining a terminal as a SerialNET connection is the naming of the terminal. So the SerialNET software can locate the connection you wish to use, it requires the terminal be named using the form SERnnn, where "nnn" is the network number associated with the terminal connection.

Thus, to define a terminal port for use with network 3, you would define a terminal named SER003, using a command similar to the following:

```
TRMDEF SER003,AM355=16:38400,TELTYP,100,100,100,,NULL
```

Note SerialNET requires the NULL modem driver be specified at the end of the TRMDEF command. It also requires that no line editor be installed on the port. The terminal driver is ignored by SerialNET, but should be specified as TELTYP to reduce overhead.

4.2.4 Setting Up Ersatz Devices

You will probably want to set up ersatz device names for the nodes on your SerialNET, as described in Chapter 1. By using ersatz device names, you can avoid the use of lengthy and cumbersome network address numbers.

4.2.5 Checking Out the Installation

Once you have made all of the changes to system initialization command file and the network initialization command file, you are ready to start testing. As a final checklist, make sure you have done the following:

1. Installed the AlphaNET software (including installing the PIC) and modified your system initialization command file to include MSGINI, NETSER, VTSER, and NETINI statements as appropriate.

2. Created a SER003.NIN file to properly define your system as a SerialNET node.

3. Defined a terminal connection for the SerialNET by using a TRMDEF command in your system initialization command file.

To test:

1. Boot up the system by MONTST'ing the new TEST.INI file.

2. If the boot up was successful, rename TEST.INI to AMOSL.INI or AMOS32.INI.

3. Physically attach the system to the network only after successfully booting the system.

4. Boot up the system using the new system initialization command file. Use a SET LINK command to establish the connection. Assuming both systems on the network have been set up properly, you should be able to communication between them.

5. If the network does not come up, recheck all items in this list. It is also possible for the cable or a connector to be bad.

When you have successfully completed TEST.INI, and MONTSTed it, you may rename the TEST.INI to your regular system initialization command file name. For AMOS/L systems, enter:

```
COPY AMOSL.OLD=AMOSL.INI RETURN
RENAME AMOSL.INI=TEST.INI/D RETURN
```

For AMOS/32 systems, enter:

```
COPY AMOS32.OLD=AMOS32.INI RETURN
RENAME AMOS32.INI=TEST.INI/D RETURN
```

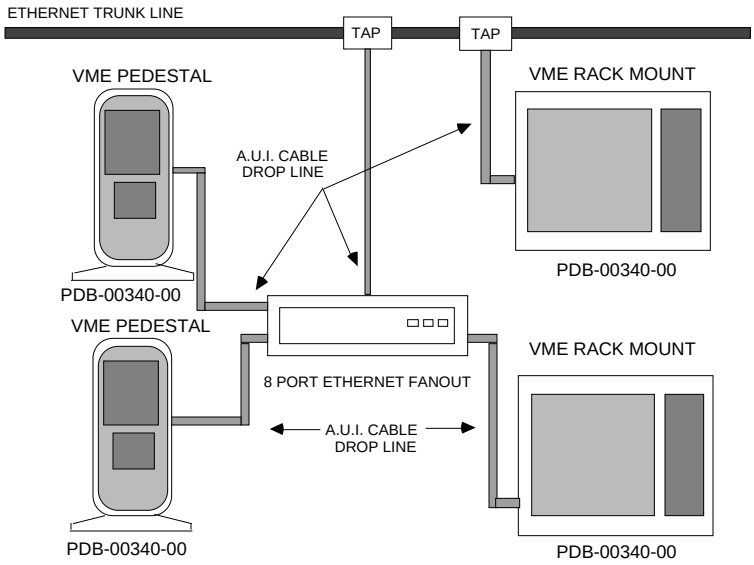
CHAPTER 5

Ethernet

Unlike networks built from existing Alpha Micro video and serial interfaces, Ethernet requires additional equipment which is not part of Alpha Micro's standard system configurations. Alpha Micro offers several different types of Ethernet controllers, with one or more models compatible with virtually every type of computer currently manufactured by Alpha Micro.

Alpha Micro's Ethernet implementation is a high speed local area network physical layer, to be used in conjunction with AlphaNET software on all Alpha Micro computers accept S-100, AM-1000, and AM-1200.

Figure 5-1 is a diagram showing several systems linked together via Alpha Micro's Ethernet. The Ethernet equipment shown interconnecting the systems in Figure 5-1 is not available from Alpha Micro, but a number of outside contractors sell and install industry standard Ethernet cabling and equipment.



TYPICAL ETHERNET INTERCONNECTION DIAGRAM

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Figure 5-1

5.1 ETHERNET INSTALLATION

Most of the instructions in this installation guide are applicable to networks using Ethernet as a communication transport device. Information on building network INI files and constructing network gateways is virtually the same for all the supported network devices, including Ethernet. However, because there are so many different types of Ethernet controllers, the instructions for installing Ethernet hardware are not included in this document.

Each type of Ethernet controller is shipped with its own set of instructions. If there are any deviations in the way the network software must be configured, those differences are pointed out in the instructions. There may also be important instructions in the release notes that accompany your AlphaNET Software. Make sure you read the release notes before you download the software.

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